



Loops: for

CS 141

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Administrivia

Common while loop format

```
//repeat body num times  
int i = 0;  
while( i < num) {  
    //body  
    i++;  
}
```

Common while loop format

<pre>//repeat body num times int i = 0; while(i < num) { //body i++; }</pre>	<pre>//new way to write this for(int i=0; i < num; i++) { //body }</pre>
---	---

for loops

Known as a *definite loop* because we know ahead of time how many times the block of statements will execute

vs.

indefinite loops (e.g. while), where we don't know how many times the block of statements will execute

for-loop elements

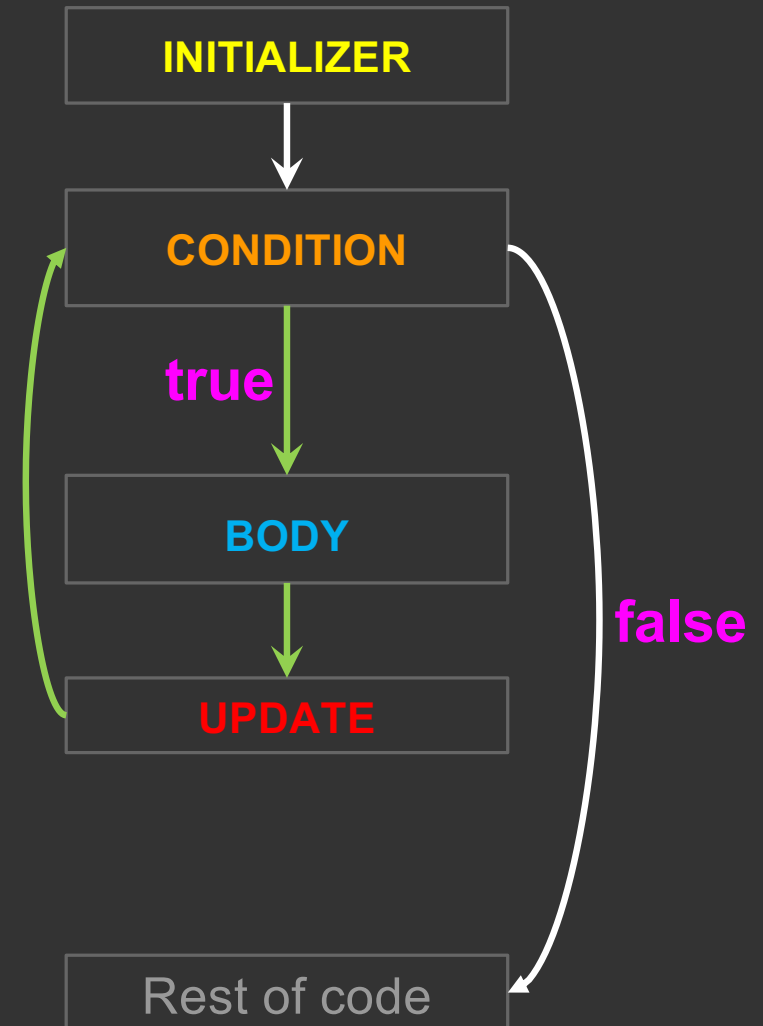
```
for ( INITIALIZER; CONDITION; UPDATE ) {  
    // ...BODY  
}  
// ...rest of code
```

INITIALIZER: happens once, at beginning

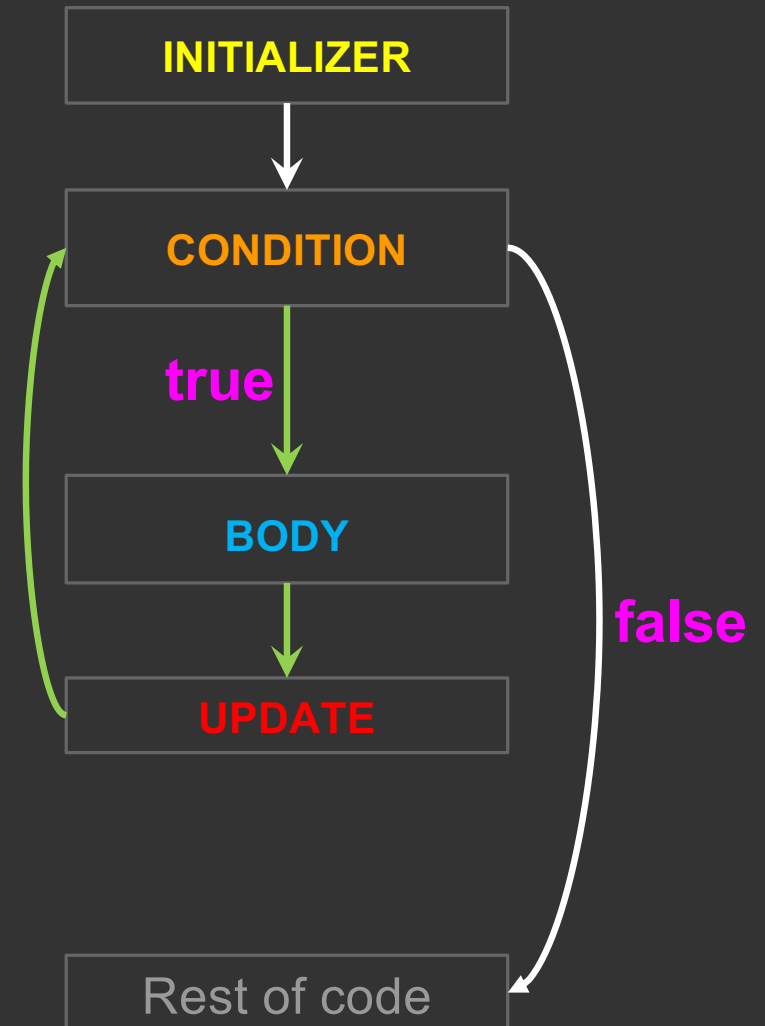
CONDITION: checked before executing body

BODY

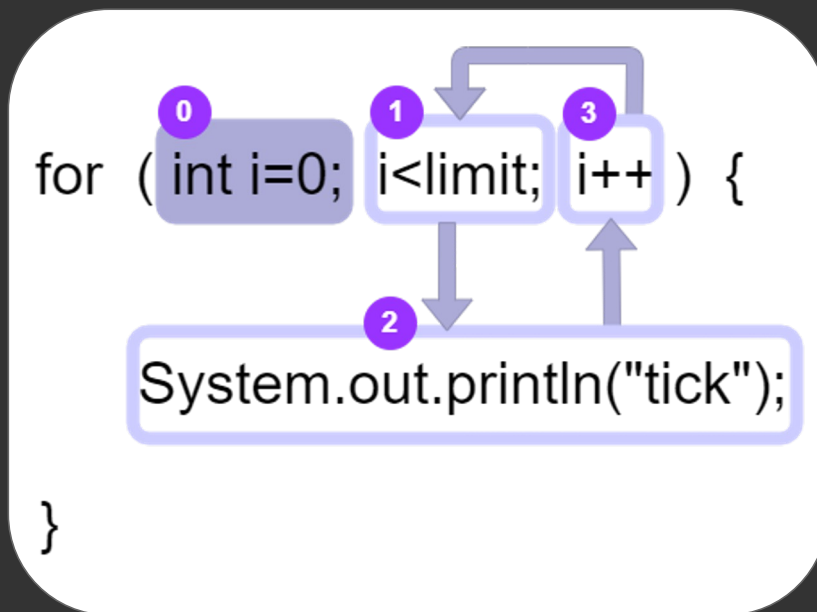
UPDATE: applied at the end of the body, then
back to condition!



```
int sum=0;  
for (int i=1; i<=5; i++) {  
    sum += i;  
}  
System.out.println(sum);
```



```
for (int i=0; i<limit; i++) {  
    System.out.println("tick");  
}
```

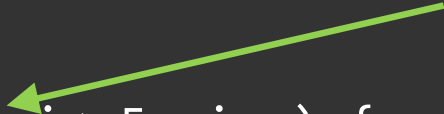


step 0: ONLY ONCE

then flow through: 1 2 3 1 2 3...
condition true?, body, increment,
condition true?, body, increment,
...
condition false? STOP LOOP.

Which part of the for loop is this?

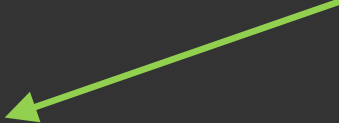
```
int sum=0;  
for (int i=1; i<=5; i++) {  
    sum += i;  
}  
System.out.println(sum);
```



- A: initializer
- B: condition
- C: update
- D: body
- E: none of the above

Which part of the for loop is this?

```
int sum=0;  
for (int i=1; i<=5; i++) {  
    sum += i;  
}  
System.out.println(sum);
```



A: initializer

B: condition

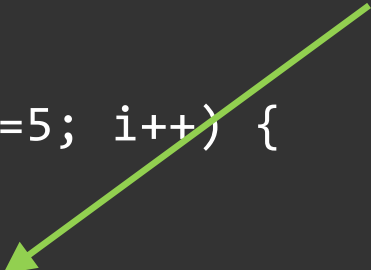
C: update

D: body

E: none of the above

Which part of the for loop is this?

```
int sum=0;  
for (int i=1; i<=5; i++) {  
    sum += i;  
}
```

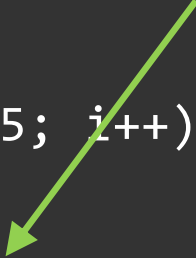


System.out.println(sum);

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- B: condition
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```
int sum=0;  
for (int i=1; i<=5; i++) {  
    sum += i;  
}
```



System.out.println(sum);

A: initializer

B: condition

C: update

D: body

E: none of the above

This is code that comes after the loop.
So it's not part of the loop itself.

**Write code to read an int and print that many
copies of X on a line**

XXXXXXXXXXXX

Write code to read an int and print that many copies of X on a line

```
Scanner sc = new Scanner(System.in);
System.out.println("enter a positive number ");
int num = sc.nextInt();
for (int i=0; i<num; i++)
{
    System.out.print("X");
}
```

Write code to read 2 ints and print a rectangle of Xs with those dimensions

```
XXXXXXXXXXXXX  
XXXXXXXXXXXXX  
XXXXXXXXXXXXX  
XXXXXXXXXXXXX
```

Write code to read 2 ints and print a rectangle of Xs with those dimensions

```
XXXXXXXXXXXXX
XXXXXXXXXXXXX
XXXXXXXXXXXXX
XXXXXXXXXXXXX
```

A nested loop is two loops,
one inside the other

```
for(int i=0; i<rows; i++) {  
    for(int j=0; j<cols; j++) {  
        ...  
    }  
}
```


Modify your code to print a hollow rectangle

```
XXXXXXXXXXXXX
X              X
X              X
XXXXXXXXXXXXX
```

Write code for different kinds of triangles

```
X
XX
XXX
XXXX
```

```
  X
 XX
XXX
XXXX
```

```
XXXX
XXX
XX
X
```

```
XXXX
XXX
XX
X
```